Talking about motion

LEARNING TARGET · K1 I can explain how position, distance, and displacement differ.

A · KEY VOCABULARY

Position	*Where are you?* Measured from a chosen zero, with a chosen + direction.
Distance	*How much travel?* Counts every part of the path. Never negative.
Displacement	*What changed from start to finish?* The sign gives the direction.
Scalar / vector	Distance has size only (scalar). Displacement has size *and* direction (vector).

B · KEY EQUATIONS + UNITS

$$\Delta x = x_f - x_i$$

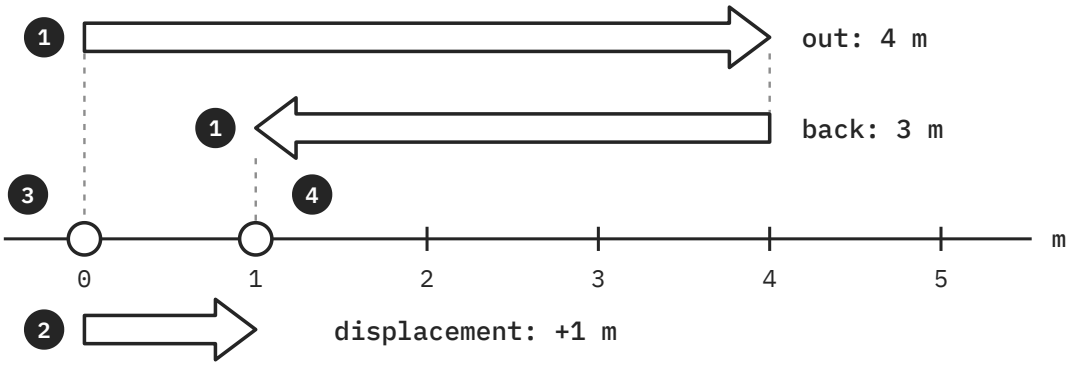
distance = leg 1 + leg 2 + …

Δ "change in" (finish − start)

x position, meters (m)

C · COLOR IT · ONE TRIP, THREE QUESTIONS

0 m → 4 m → 1 m. Right is +.



- ☐ **1 · PENCIL** Distance: trace both legs
- ☐ **2 · BLUE** Displacement arrow
- ☐ **3 · BLUE** Start position
- ☐ **4 · BLUE** Finish position

D · WORKED EXAMPLE

A cart goes from 2 m to 6 m, then back to 4 m. Right is positive. Find the distance and the displacement.

GIVEN	EQUATION	WORK	ANSWER
x_i = 2 m turn at 6 m x_f = 4 m	distance = leg 1 + leg 2 $\Delta x = x_f - x_i$	4 m + 2 m 4 m − 2 m	distance **6 m** displacement **+2 m**

E · THE 2026-XJ THREAD

DAY 2 OF 22 · NEW TOOL: A REFERENCE FRAME

Before we can say where XJ is, we need a zero and a + direction. Analysts measure its position from Earth and track its displacement, not the winding path it travels.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else